

Exercises 8.C

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Exercise 1. Suppose $T \in \mathcal{L}(\mathbb{C}^4)$ is such that the eigenvalues of T are 3, 5, 8. Prove that $(T - 3I)^2(T - 5I)^2(T - 8I)^2 = 0$.

Proof. Since $\dim \mathbb{C}^4 = 4$, the degree of the minimal polynomial p of T is at most 4. The degree of the above polynomial, call it q , is 6. Since $p(T) = 0$, the zeroes of p are exactly the eigenvalues of T , namely 3, 5, 8. This means that p can be written as

$$p(z) = (z - 3)^{n_1}(z - 5)^{n_2}(z - 8)^{n_3}$$

where $n_1, n_2, n_3 \in \{1, 2\}$. No matter what the values of those powers end up being, each is at most 2, so q is clearly a polynomial multiple of p . By 8.46, this means $q(T) = 0$. $\square$

Exercise 2. Suppose V is a complex vector space. Suppose $T \in \mathcal{L}(V)$ is such that 5 and 6 are eigenvalues of T and that T has no other eigenvalues. Prove that $(T - 5I)^{n-1}(T - 6I)^{n-1} = 0$, where $n = \dim V$.

Proof. We know that the minimal polynomial, p , of T is of degree at most $\dim V$. We can also see that given that T has two distinct eigenvalues, $\dim V \geq 2$, and the multiplicity of each eigenvalue is at most $n - 1$ since their multiplicities must each be at least 1 and their sum must equal n because V is a complex vector space. Thus, the characteristic polynomial q can be described as

$$q(z) = (z - 5)^{m_1}(z - 6)^{m_2}$$

where $m_1, m_2 \in \{1, \dots, n - 1\}$. Given this, clearly the polynomial in question (call it r) is a polynomial multiple of q , and since $q(T) = 0$, so is r . $\square$

Exercise 3. Give an example of an operator on $\mathbb{C}^4$ whose characteristic polynomial equals $(z - 7)^2(z - 8)^2$.

Proof. The operator $T \in \mathcal{L}(V)$ defined by

$$T(z_1, z_2, z_3, z_4) = (7z_1, 7z_2, 8z_3, 8z_4)$$

clearly has as its eigenvalues 7 and 8, with the corresponding eigenspaces each being two-dimensional. Since that makes the sum of the dimension of the eigenspaces 4 (which is $\dim V$), the dimension of the generalized eigenspaces

must also each be 2, and thus the multiplicity of each eigenvalue is 2. It follows by definition that the characteristic polynomial of T is $(z - 7)^2(z - 8)^2$ as desired. $\square$

Exercise 8. Suppose $T \in \mathcal{L}(V)$. Prove that T is invertible if and only if the constant term in the minimal polynomial of T is nonzero.

Proof. Denote the minimal polynomial of T as $p(z) = a_0 + a_1z + \cdots + a_nz^n$. First, assume that $a_0 \neq 0$. Then

$$\begin{aligned} a_0I &= -a_1T + \cdots + a_nT^n \\ \implies I &= -\frac{a_1}{a_0}T + \cdots + \frac{a_n}{a_0}T^n \\ &= T\left(-\frac{a_1}{a_0}I + \cdots + \frac{a_n}{a_0}T^{n-1}\right), \end{aligned}$$

so T is invertible.

Now assume that $a_0 = 0$. Then we have

$$0 = a_1T + \cdots + a_nT^n \quad (1)$$

$$= T(a_1I + \cdots + a_nT^{n-1}). \quad (2)$$

For convenience, we define $q \in \mathcal{P}(\mathbf{F})$ by $q(z) = a_1 + \cdots + a_nz^{n-1}$ so that the right side of (2) becomes $T(q(T))$. Since q is a polynomial of degree less than n , q cannot be the zero map. Thus there exists some $v \in V$ for which $(q(T))v \neq 0$. Equation (2) indicates that $(q(T))v \in \text{null } T$, so T is not invertible. $\square$

Exercise 12. Suppose V is a complex vector space and $T \in \mathcal{L}(V)$. Prove that V has a basis consisting of eigenvectors of T if and only if the minimal polynomial of T has no repeated zeroes.

Proof. We begin with the easy direction. If V has a basis consisting of eigenvectors of T , then given eigenvalues $\lambda_0, \dots, \lambda_n$, the polynomial

$$p(T) = (T - \lambda_0I) \cdots (T - \lambda_nI)$$

sends every basis element to zero, and thus is the zero map. p must then be the minimal polynomial because taking away any factors would take away zeroes.

For the other direction, assume that the minimal polynomial of T has no repeated zeroes. Then write the minimal polynomial p as

$$p(T) = (T - \lambda_0) \cdots (T - \lambda_n)$$

where each λ_j is a distinct zero of p . Since p is the minimal polynomial of T , p sends each basis element to 0. $\square$